A SECOND STRONGLY REGULAR GRAPH WITH PARAMETERS $(243, 22, 1, 2)$

DILAN DJEMATENE

Abstract. We construct a strongly regular graph with parameters $(243, 22, 1, 2)$ that is not isomorphic to the Berlekamp–van Lint–Seidel graph. This answers the uniqueness question for these parameters negatively. The new graph has an automorphism group of order 4320 with orbits of sizes $3, 60, 180$, and it contains 405 induced 3×3 grids, against 1485 for the Berlekamp–van Lint–Seidel graph.

It is obtained by replacing the subgraph induced on the 180 vertices at distance 2 from a triangle by the second solution of the linear equations that this subgraph must satisfy. Unlike the two previously known graphs with $\lambda = 1$, $\mu = 2$, the new graph violates Makhnev's grid condition. Consequently it is a counterexample to Reimbayev's conjecture on the number of hexagons in strongly regular graphs with $\lambda = 1$, $\mu = 2$: it has 5 000 130 induced hexagons, against the conjectured 4 980 690.

1. Introduction

A strongly regular graph with $\lambda = 1$ and $\mu = 2$ has $v = 1 + k^2/2$. Integrality of the eigenvalue multiplicities leaves only $k \in \{4, 14, 22, 112, 994\}$, that is $v \in \{9, 99, 243, 6273, 494019\}$. For $v = 9$ the graph is the 3×3 grid. For $v = 99$, existence is Conway's well-known problem. For $v = 243$ the only known example was the coset graph of the ternary Golay code, constructed by Berlekamp, van Lint and Seidel [1] (*BvLS* below). Its uniqueness is listed as open in [3, 2, 7]; it is known only among rank 3 graphs, abelian Cayley graphs and translation graphs [3].

Theorem 1. *There is a strongly regular graph Γ' with parameters $(243, 22, 1, 2)$ that is not isomorphic to BvLS. Its automorphism group has order 4320, with vertex orbits of sizes 3, 60 and 180.*

The graph is given explicitly in the ancillary files, in graph6 format (`srg243_new.g6`) and as an edge list (`srg243_new.edges`). A self-contained verifier in pure Python (`verify_srg243.py`) checks 22-regularity and the values $\lambda = 1$, $\mu = 2$ on all 29 403 pairs of vertices.

2. The construction

Let Γ be an $\mathrm{srg}(v, k, 1, 2)$ and $T = \{x, y, z\}$ a triangle. Every vertex outside T is adjacent to at most one vertex of T, by $\lambda = 1$. Let C be the

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$3(k-2)$ vertices adjacent to T, and let F be the remaining *far* vertices. A far vertex w is at distance 2 from each vertex of T, so it has exactly two neighbours in each of $N(x)\setminus T$, $N(y)\setminus T$ and $N(z)\setminus T$, hence 6 neighbours in C and $k-6$ in F.

Counting common neighbours of $u \in C$ and $w \in F$ gives the linear equations

$$(1)\quad \#\{w' \in N_F(w) : w' \sim u\} = (2 - A_{uw}) - |N_C(u) \cap N_C(w)| \qquad (u \in C).$$

In each far vertex w, these equations constrain the set $N_F(w)$ of far neighbours of w, given the adjacencies between C and F.

Lemma 2. *Let Γ be BvLS and T any triangle. Then $|C| = 60$ and $|F| = 180$, and for every far vertex w the system (1), together with $|N_F(w)| = 16$ and $w \notin N_F(w)$, has exactly two solutions: the true far neighbourhood of w, and a second one that differs from it in 8 of its 16 elements.*

The lemma is a finite computation: for each of the 180 rows we enumerated all solutions of (1) exhaustively with an exact integer constraint solver (OR-Tools CP-SAT). BvLS is arc-transitive, so all triangles are equivalent.

Write F' for the relation in which each far vertex is joined to its *second* solution. The relation F' is symmetric, and changing to it removes 720 edges inside F and adds 720 others. Let Γ' be the graph obtained from BvLS by replacing the subgraph induced on F by F'; all adjacencies involving $T \cup C$ are kept.

Proof of Theorem 1. Γ' *is an* $\mathrm{srg}(243, 22, 1, 2)$. Regularity is clear, since every row of F' has 16 elements. For a pair $\{u, w\}$ with $u \in C$ and $w \in F$, the number of common neighbours is preserved by (1). For the other pairs meeting $T \cup C$, every edge from the pair to a common neighbour has an end in $T \cup C$, so these counts are unchanged. For the pairs inside F the values $\lambda = 1$, $\mu = 2$ are checked by direct computation. The ancillary verifier checks all 29 403 pairs.

Γ' *is not isomorphic to BvLS.* For a vertex p, let $g(p)$ be the number of induced 3×3 grids ($K_3 \times K_3$) through p; this is an isomorphism invariant. In BvLS, $g \equiv 55$ (there are 1485 grids). In Γ', g takes the values 55, 19, 13 on 3, 60, 180 vertices respectively, for a total of 405 grids. The verifier computes these numbers.

Automorphisms. nauty [5] gives $|\mathrm{Aut}\,\Gamma'| = 4320$ with orbits of sizes $3, 60, 180$, namely T, C and F; and $|\mathrm{Aut}\,\mathrm{BvLS}| = 3\,849\,120 = 3^5 \cdot 2 \cdot |M_{11}|$, as expected. □

3. Twisting, and Reimbayev's conjecture

Call the triangles of a graph with $\lambda = 1$ its *lines*. Makhnev [4] considers the condition

> $(*)$ two disjoint lines joined by at least two edges are joined by exactly three edges.

For $\mu = 2$, $(*)$ is equivalent to: every two intersecting lines lie in a 3×3 grid. Makhnev proved that no $\mathrm{srg}(99, 14, 1, 2)$ satisfies $(*)$. Both previously known graphs of the family satisfy it.

Proposition 3. *In Γ', exactly 19 440 pairs of disjoint lines are joined by exactly two edges. Hence Γ' violates $(*)$. Moreover Γ' has 2430 pairs of lines joined by three edges (prisms), against 8910 in BvLS.*

Reimbayev [6] proved that an $\mathrm{srg}(n, k, 1, 2)$ has at least

$$H_0 = \tfrac{1}{12}\, n\, k\, (k-2)\, (2k^2 - 21k + 53)$$

induced hexagons, with equality exactly when $(*)$ holds (his quantity n_3 vanishes). He conjectured that every strongly regular graph with $\lambda = 1$, $\mu = 2$ attains this bound [6, §4].

Corollary 4. *Reimbayev's conjecture is false. For $n = 243$, $k = 22$ we have $H_0 = 4\,980\,690$; BvLS has exactly H_0 induced hexagons, while Γ' has $5\,000\,130 = H_0 + 19\,440$.*

The hexagon counts were obtained by direct enumeration (ancillary program `hexagons.c`). The program was checked against brute force on random graphs and on the 3×3 grid, which has $6 = H_0$ induced hexagons.

Remark 5. In the language of [4], Γ' is the first known *twisted* member of the family. Its twisting is concentrated away from the original triangle: the three vertices of T have $g = 55 = \binom{11}{2}$, so every two lines through each of them lie in a grid.

4. Switching classes

Applying the same construction to every triangle of Γ' gives Table 1. The triangles are classified by the multiset of the values g of their vertices.

Table 1. The far-graph switching on the triangles of Γ' (one triangle of each type).

type	number	solutions of (1) that give a strongly regular graph
$(55, 55, 55)$	1	Γ' and BvLS
$(19, 19, 55)$	30	Γ' only
$(19, 19, 19)$	20	Γ' only
$(13, 13, 19)$	540	Γ' only
$(13, 13, 13)$	300	Γ' only

So the class of BvLS under this switching, applied to one triangle at a time, is exactly $\{\mathrm{BvLS}, \Gamma'\}$.

The construction is not a Godsil–McKay switching. In fact, for the relevant parameters, no Godsil–McKay switching is possible at all.

Proposition 6. *Let* Γ *be an* $\mathrm{srg}(v,k,1,2)$ *with* $k \in \{14, 22, 112, 994\}$. *Then* Γ *has no Godsil–McKay switching set other than* $\emptyset$ *and* V.

Proof. Let C be a switching set: $|C| = 2n$, C induces a d-regular graph, and every vertex outside C has 0, n or $2n$ neighbours in C. For $n = 1$ the switching is a relabelling, so assume $n \geq 2$. Fix $a \in C$. Each of its $k-d$ outside neighbours has at least $n-1$ neighbours in $C \setminus \{a\}$. A vertex $x \in C \setminus \{a\}$ has at most 1 common neighbour with a if $x \sim a$, and at most 2 otherwise. Counting pairs therefore gives

$$(k-d)(n-1) \leq d+2(2n-1-d), \qquad \text{that is} \qquad (k-4)n-(k-2) \leq d(n-2).$$

For $n = 2$ this gives $k \leq 6$, a contradiction. For $n \geq 3$ it gives $d \geq k-3$, so at most $6n$ edges leave C. Since Γ is k-regular with second largest eigenvalue r (namely $r = 3, 4, 10, 31$ for $k = 14, 22, 112, 994$), the number of edges leaving C is at least $(k-r)\,2n\,(v-2n)/v$. Hence $2n \geq v\,(1-3/(k-r))$, which gives $n \geq 36$, 102, 3045 and an even larger value for $k = 14$, 22, 112 and 994 respectively. In every case $n > k$. Then no outside vertex can have n or $2n$ neighbours in C, so no edge leaves C. Since Γ is connected, $C = V$. □

Remark 7 (Data and reproducibility). Ancillary files:

- the new graph (`srg243_new.g6` and `.edges`);
- BvLS (`bvls243.edges`);
- the verifier `verify_srg243.py` (regularity, λ, μ, grids);
- `hexagons.c`.

The automorphism groups were computed with nauty.

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